EECT/ CE 6325 VLSI Design

PROJECT #3: INVERTER DESIGN AND LAYOUT

Fall 2026

Due: **Friday Oct 16**

Project Introduction

For this project you will be using the TSMC 16nm FinFET process and Cadence Design tools to design, layout and characterize an inverter. This project will get you familiar with Cadence and also with the simulation of the layouts that you create.

Project Description/Requirements

1) Review (carefully) the inverter layout tutorial for the TSMC 16nm FinFET process. (TSMCN16 Process.pdf)
2) The input and output pin labels must be capitalized in both schematic view and layout view: IN, OUT
3) The power and ground pin labels in layout should be VDD and GND
4) There will be 4 fins for both the pMOS and nMOS devices.
5) Simulate the inverter (tutorial: TSMCN16 Process.pdf & HSPICE - VLSI tutorial.pdf)
6) The input slew rate is 20ps (the slew rate, for this problem, is defined as the time for the input to go from low (0.2* Vdd) to high (0.8* Vdd) and vice versa). Use a piecewise linear input waveform or, if you dare, a pulse waveform accordingly.
7) VDD is 0.8V & GND is 0V.
8) ***DESIGN OBJECTIVE:*** Minimize the bounding box area (H*W) of your inverter layout.

Report Layout

1) Submit only a soft copy. Because of that, black backgrounds for waveforms are encouraged to enhance readability.
2) A cover page containing:
   - Student names, NetID and project titles
   - Clearly state your rise delay, fall delay, output rise edge rate (20% to 80%), and output fall edge rate (80% to 20%) and the area (widths & lengths are measured from highest Metal1 layer to the lowest Metal1 layer) on the front.
3) Soft copy of the report should include:
   - Spice test setup file
   - Waveforms showing how you measured the above rise/fall delays and output edge rates.
   - Inverter layout with rulers (**from cadence**)
   - Extracted Spice Netlist
4) Upload the zipped folder containing your inverter layout, schematic on eLearning

Grading Breakdown

75% Proper layout according to given parameters; functional correctness of the inverter; proper measurement of characteristics.

25% Report clarity and completeness